

APACHE STORM

Introduction:

Apache Storm is a free and open-source distributed real-time computation system designed for processing streams of data in real-time. Originally developed by Nathan Marz at BackType and later acquired by Twitter, Storm has become one of the most popular frameworks for stream processing in the big data ecosystem.

What is Apache Storm?

Apache Storm is a distributed, fault-tolerant, and real-time stream processing framework that can process millions of tuples per second. It provides a simple programming model for distributed computing, making it easy to process unbounded streams of data reliably.

Key Characteristics:

- **Real-time processing:** Processes data as it arrives
- **Distributed:** Runs on clusters of machines
- **Fault-tolerant:** Automatically handles failures
- **Scalable:** Can handle increasing data volumes

Stream Processing Fundamentals

Stream processing is a computing paradigm that processes data continuously as it flows through the system, rather than storing it first and processing it later (batch processing). This approach enables real-time analytics and immediate responses to events.

Core Storm Concepts

Streams

A stream is an unbounded sequence of tuples (data records) that are processed in parallel in a distributed fashion. Each tuple consists of a list of named fields.

Spouts

Spouts are the sources of streams in a Storm topology. They read data from external sources (databases, message queues, APIs) and emit tuples into the topology.

Bolts

Bolts are the processing units in Storm. They consume input streams, process the data, and optionally emit new streams. Bolts can perform various operations like filtering, aggregation, database operations, and more.

Topologies

A topology is a graph of spouts and bolts connected by stream groupings. It represents the overall data processing logic and runs indefinitely until explicitly terminated.

Stream Groupings

Stream groupings define how tuples are distributed among bolt instances:

- **Shuffle Grouping:** Random distribution
- **Fields Grouping:** Based on field values
- **All Grouping:** Replicated to all instances
- **Global Grouping:** All tuples go to one instance
- **None Grouping:** Same as shuffle grouping
- **Direct Grouping:** Producer decides the consumer

Processing Guarantees

Storm provides different levels of message processing guarantees:

At-least-once Processing

Every tuple is guaranteed to be processed at least once. This is achieved through Storm's acknowledgment mechanism.

Exactly-once Processing

With Storm Trident, you can achieve exactly-once processing semantics, ensuring each tuple is processed exactly once.

Architecture and Components

Storm Cluster Architecture

A Storm cluster consists of several key components:

Nimbus (Master Node)

- Distributes code around the cluster
 - Assigns tasks to workers
 - Monitors for failures and reassigns tasks
- Similar to Hadoop's JobTracker

Supervisor (Worker Nodes)

- Listens for work assigned to its machine
- Starts and stops worker processes
- Runs on each worker machine

Worker Processes

- Execute a subset of a topology
- Run multiple threads called

executors • Each

executor runs

multiple tasks

Apache Storm vs Apache Spark Streaming

Feature	Apache Storm	Apache Spark Streaming
Processing Model	True streaming	Micro-batching
Latency	Sub-second	1-2 seconds
Throughput	Moderate	High
Fault Tolerance	Record-level	Batch-level
State Management	Limited	Rich support
Learning Curve	Moderate	Steep

Apache Storm vs Apache Flink

Feature	Apache Storm	Apache Flink
Processing Guarantee	At-least-once	Exactly-once
Windowing	Basic	Advanced
State Management	Limited	Rich
Performance	Good	Excellent
Maturity	High	Growing

